as $d\mathbf{s} = -\hat{\mathbf{x}}\, dy\, dz$. Hence,

$$L_{12} = \frac{1}{I}\int_{y=0.05}^{0.20}\int_{z=0.1}^{0.4}\left(-\hat{\mathbf{x}}\,\frac{\mu_0 I}{2\pi y}\right)\cdot(-\hat{\mathbf{x}}\,dy\,dz)$$
$$= \frac{\mu_0}{2\pi}\left(\int_{0.05}^{0.20}\frac{dy}{y}\right)\left(\int_{0.1}^{0.4}dz\right)$$
$$= \frac{0.3\mu_0}{2\pi}\,(\ln y)\Big|_{0.05}^{0.20} = \frac{0.3\mu_0}{2\pi}\ln\left(\frac{0.2}{0.05}\right) = 83\text{ nH}.$$

Concept Question 5-15: What is the magnetic field like in the interior of a long solenoid?

Concept Question 5-16: What is the difference between self-inductance and mutual inductance?

Concept Question 5-17: How is the inductance of a solenoid related to its number of turns N?

5-8 Magnetic Energy

When we introduced electrostatic energy in Section 4-10, we did so by examining what happens to the energy expended in charging up a capacitor from zero voltage to some final voltage V. We introduce the concept of magnetic energy by considering an inductor with inductance L connected to a current source. Suppose that we were to increase the current i flowing through the inductor from zero to a final value I. From circuit theory, we know that the instantaneous voltage v across the inductor is given by $v = L\, di/dt$. We will derive this relationship from Maxwell's equations in Chapter 6, thereby justifying the use of the i–v relationship for the inductor. Power p equals the product of v and i, and the time integral of power is work, or energy. Hence, the total energy in joules (J) expended in building up a current I in the inductor is

$$W_m = \int p\, dt = \int iv\, dt = L\int_0^I i\, di = \frac{1}{2}LI^2 \quad \text{(J)}. \qquad (5.103)$$

We call this the ***magnetic energy*** stored in the inductor.

To justify this association, consider the solenoid inductor. Its inductance is given by Eq. (5.95) as $L = \mu N^2 S/l$, and the magnitude of the magnetic flux density in its interior is given by Eq. (5.90) as $B = \mu NI/l$, implying that $I = Bl/(\mu N)$. Using these expressions for L and I in Eq. (5.103), we obtain

$$W_m = \frac{1}{2}LI^2 = \frac{1}{2}\left(\mu\,\frac{N^2}{l}S\right)\left(\frac{Bl}{\mu N}\right)^2$$
$$= \frac{1}{2}\,\frac{B^2}{\mu}\,(lS) = \frac{1}{2}\,\mu H^2 v, \qquad (5.104)$$

where $v = lS$ is the volume of the interior of the solenoid and $H = B/\mu$. The expression for W_m suggests that the energy expended in building up the current in the inductor is stored in the magnetic field with ***magnetic energy density*** w_m, defined as the magnetic energy W_m per unit volume,

$$w_m = \frac{W_m}{v} = \frac{1}{2}\mu H^2 \qquad (\text{J/m}^3). \qquad (5.105)$$

▶ Even though this expression was derived for a solenoid, it remains valid for any medium with a magnetic field **H**. ◀

Furthermore, for any volume v containing a material with permeability μ (including free space with permeability μ_0), the total magnetic energy stored in a magnetic field **H** is

$$W_m = \frac{1}{2}\int_v \mu H^2\, dv \qquad \text{(J)}. \qquad (5.106)$$

Example 5-10: Magnetic Energy in a Coaxial Cable

Derive an expression for the magnetic energy stored in a coaxial cable of length l and inner and outer radii a and b. The current flowing through the cable is I and its insulation material has permeability μ.

Solution: From Eq. (5.97), the magnitude of the magnetic field in the insulating material is

$$H = \frac{B}{\mu} = \frac{I}{2\pi r},$$

where r is the radial distance from the center of the inner conductor (**Fig. 5-29**). The magnetic energy stored in the coaxial cable therefore is

$$W_m = \frac{1}{2}\int_v \mu H^2\, dv = \frac{\mu I^2}{8\pi^2}\int_v \frac{1}{r^2}\, dv.$$

Since H is a function of r only, we choose dv to be a cylindrical shell of length l, radius r, and thickness dr along the radial direction. Thus, $dv = 2\pi rl\, dr$ and

$$W_m = \frac{\mu I^2}{8\pi^2}\int_a^b \frac{1}{r^2}\cdot 2\pi rl\, dr = \frac{\mu I^2 l}{4\pi}\ln\left(\frac{b}{a}\right) = \frac{1}{2}LI^2 \quad \text{(J)}$$

with L given by Eq. (5.99).

作为 ds = −x̂ dy dz。因此，

$$L_{12} = \frac{1}{I}\int_{y=0.05}^{0.20}\int_{z=0.1}^{0.4}\left(-\hat{\mathbf{x}}\,\frac{\mu_0 I}{2\pi y}\right)\cdot(-\hat{\mathbf{x}}\,dy\,dz)$$
$$= \frac{\mu_0}{2\pi}\left(\int_{0.05}^{0.20}\frac{dy}{y}\right)\left(\int_{0.1}^{0.4}dz\right)$$
$$= \frac{0.3\mu_0}{2\pi}\,(\ln y)\Big|_{0.05}^{0.20} = \frac{0.3\mu_0}{2\pi}\ln\left(\frac{0.2}{0.05}\right) = 83\text{ nH}.$$

概念问题 5-15：长螺线管内部的磁场是什么样的？

概念问题 5-16：自感和互感有什么区别？

概念问题 5-17：螺线管的电感与其匝数 N 有何关系？

5-8 磁能

当我们在第 4-10 节中引入静电能时，我们通过检查将电容器从零电压充电到某个最终电压 V 所消耗的能量发生了什么来做到这一点。我们通过考虑连接到电流源的电感 L 的电感器来引入磁能的概念。假设我们要将流过电感器的电流 i 从零增加到最终值 I。根据电路理论，我们知道电感器上的瞬时电压 υ 由 υ = L di/dt 给出。我们将从第 6 章中的麦克斯韦方程组中推导出这个关系，从而证明对电感器使用 i–υ 关系是合理的。功率p等于υ和i的乘积，功率的时间积分就是功，或者说能量。因此，在电感器中建立电流 I 所消耗的总能量（以焦耳 (J) 为单位）为

$$W_m = \int p\, dt = \int iv\, dt = L\int_0^I i\, di = \frac{1}{2}LI^2 \quad \text{(J)}. \qquad (5.103)$$

我们称之为存储在电感器中的磁能。

为了证明这种关联的合理性，请考虑螺线管电感器。其电感由方程式给出。 (5.95) 为 L = μ N²S/l，其内部磁通密度的大小由方程 (5.95) 给出。 (5.90) 为 B = μ NI/l，意味着 I = Bl/(μ N)。在等式中使用这些表达式表示 L 和 I。 （5.103），我们得到

$$W_m = \frac{1}{2}LI^2 = \frac{1}{2}\left(\mu\,\frac{N^2}{l}S\right)\left(\frac{Bl}{\mu N}\right)^2$$
$$= \frac{1}{2}\,\frac{B^2}{\mu}\,(lS) = \frac{1}{2}\,\mu H^2 v, \qquad (5.104)$$

其中，υ = lS 是螺线管内部的体积，H = B/μ 。 W_m 的表达式表明，在电感器中建立电流所消耗的能量存储在磁能密度为 w_m 的磁场中，定义为每单位体积的磁能 W_m ，

$$w_m = \frac{W_m}{v} = \frac{1}{2}\mu H^2 \qquad (\text{J/m}^3). \qquad (5.105)$$

▶ 尽管这个表达式是针对螺线管导出的，但它对于任何具有磁场 H 的介质仍然有效。◀

此外，对于包含具有磁导率 μ（ 的材料（包括具有磁导率 μ_0) 的自由空间）的任何体积 υ，存储在磁场 H 中的总磁能为

$$W_m = \frac{1}{2}\int_v \mu H^2\, dv \qquad \text{(J)}. \qquad (5.106)$$

例 5-10：同轴电缆中的磁能

导出长度为 l、内半径和外半径为 a 和 b 的同轴电缆中存储的磁能的表达式。流过电缆的电流为I，其绝缘材料具有磁导率μ 。

解决方案：从方程。 （5.97），绝缘材料中的磁场强度为

$$H = \frac{B}{\mu} = \frac{I}{2\pi r},$$

其中r是距内导体中心的径向距离（图5-29）。因此，同轴电缆中存储的磁能为

$$W_m = \frac{1}{2}\int_v \mu H^2\, dv = \frac{\mu I^2}{8\pi^2}\int_v \frac{1}{r^2}\, dv.$$

由于 H 仅是 r 的函数，因此我们选择 dυ 作为沿径向长度为 l、半径为 r、厚度为 dr 的圆柱壳。因此，dυ = 2π rl dr 和

$$W_m = \frac{\mu I^2}{8\pi^2}\int_a^b \frac{1}{r^2}\cdot 2\pi rl\, dr = \frac{\mu I^2 l}{4\pi}\ln\left(\frac{b}{a}\right) = \frac{1}{2}LI^2 \quad \text{(J)}$$

L 由方程给出。 （5.99）。

Technology Brief 11: Inductive Sensors

Magnetic coupling between different coils forms the basis of several different types of inductive sensors. Applications include the measurement of position and displacement (with submillimeter resolution) in device fabrication processes, ***proximity detection*** of conductive objects, and other related applications.

Linear Variable Differential Transformer (LVDT)

► An LVDT comprises a primary coil connected to an ac source (typically a sine wave at a frequency in the 1–10 kHz range) and a pair of secondary coils, all sharing a common ***ferromagnetic core*** (**Fig. TF11-1**). ◄

The magnetic core serves to couple the magnetic flux generated by the primary coil into the two secondaries, thereby inducing an output voltage across each of them. The secondary coils are connected in opposition, so when the core is positioned at the magnetic center of the LVDT, the individual output signals of the secondaries cancel each other out, producing a null output voltage. The core is connected to the outside world via a nonmagnetic push rod. When the rod moves the core away from the magnetic center, the magnetic fluxes induced in the secondary coils are no longer equal, resulting in a nonzero output voltage. The LVDT is called a "linear" transformer because the amplitude of the output voltage is a linear function of displacement over a wide operating range (**Fig. TF11-2**).

The cutaway view of the LVDT model in **Fig. TF11-3** depicts a configuration in which all three coils—with the primary straddled by the secondaries—are wound around a glass tube that contains the magnetic core and attached rod. Sample applications are illustrated in **Fig. TF11-4**.

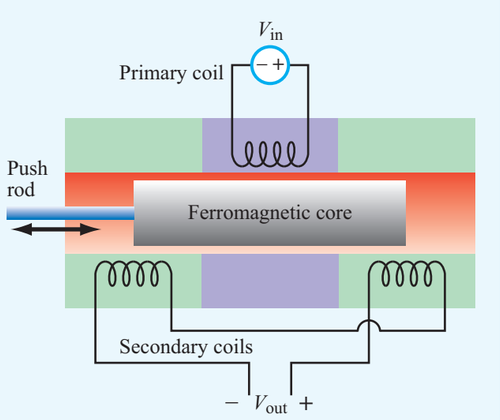


Figure TF11-1 Linear variable differential transformer (LVDT) circuit.

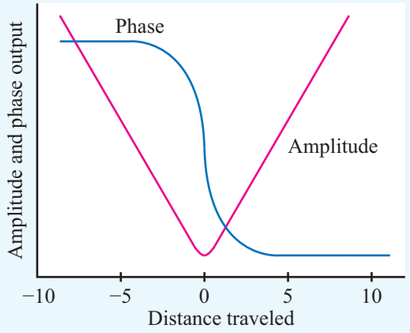


Figure TF11-2 Amplitude and phase responses as functions of the distance by which the magnetic core is moved away from the center position.

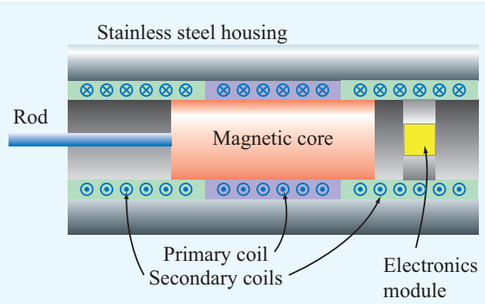


Figure TF11-3 Cutaway view of LVDT.

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技术简介 11：电感式传感器

不同线圈之间的磁耦合构成了几种不同类型的电感式传感器的基础。应用包括设备制造过程中的位置和位移测量（亚毫米分辨率）、导电物体的接近检测以及其他相关应用。

线性可变差动变压器 (LVDT)

► LVDT 包括一个连接到交流电源（通常是频率在 1 -10 kHz 范围内的正弦波）的初级线圈和一对次级线圈，所有线圈共享一个公共铁磁芯（图 TF11-1）。 ◄

磁芯用于将初级线圈产生的磁通量耦合到两个次级线圈中，从而在每个次级线圈上感应出输出电压。次级线圈反向连接，因此当磁芯位于 LVDT 的磁中心时，次级线圈的各个输出信号相互抵消，产生零输出电压。核心通过非磁性推杆与外界相连。当棒将磁芯移离磁中心时，会产生感应磁通量

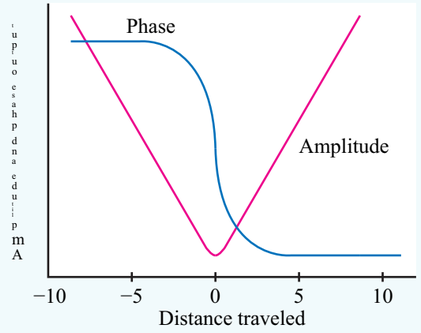
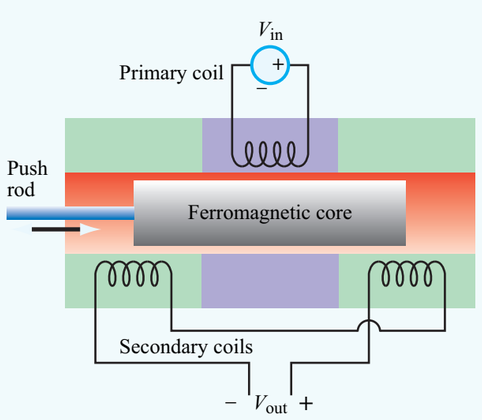


图 TF11-2 幅度和相位响应作为磁芯远离中心位置的距离的函数。

次级线圈中的电压不再相等，导致输出电压非零。 LVDT 被称为“线性”变压器，因为输出电压的幅度是宽工作范围内位移的线性函数（图 TF11-2）。

图 TF11-3 中 LVDT 模型的剖面图描绘了一种配置，其中所有三个线圈（初级线圈跨在次级线圈上）都缠绕在包含磁芯和附加杆的玻璃管上。示例应用如图 TF11-4 所示。



图TF11-1 线性可变差动变压器（LVDT）电路。

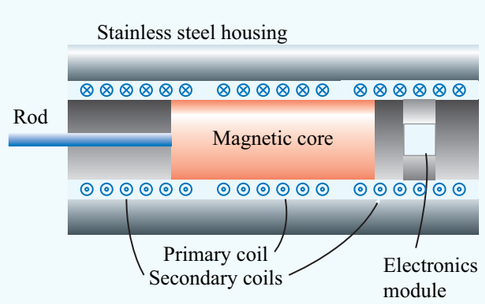


图 TF11-3 LVDT 剖面图。

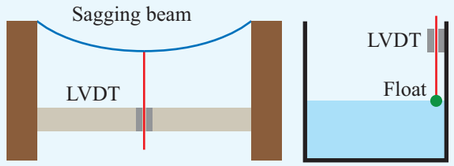


Figure TF11-4 LVDT for measuring beam deflection and as a fluid-level gauge.

Eddy-Current Proximity Sensor

The transformer principle can be applied to build a proximity sensor in which the output voltage of the secondary coil becomes a sensitive indicator of the presence of a conductive object in its immediate vicinity (**Fig. TF11-5**).

► When an object is placed in front of the secondary coil, the magnetic field of the coil induces eddy (circular) currents in the object that generate magnetic fields of their own having a direction that opposes the magnetic field of the secondary coil. ◄

The reduction in magnetic flux causes a drop in output voltage with the magnitude of the change being dependent on the conductive properties of the object and its distance from the sensor.

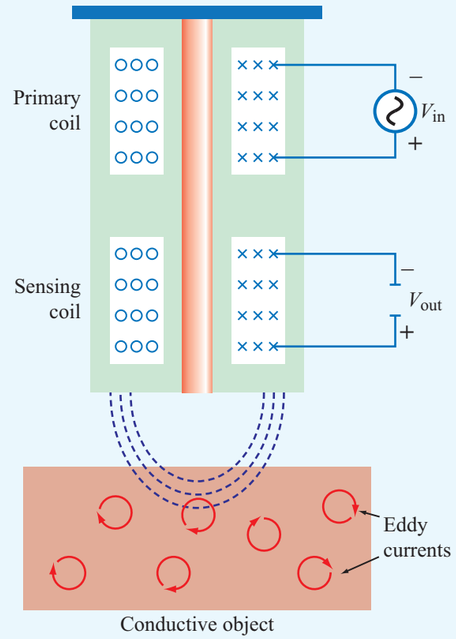


Figure TF11-5 Eddy-current proximity sensor.

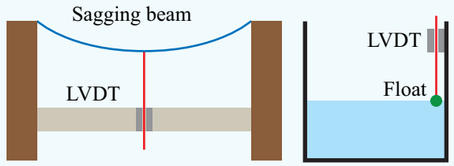


图 TF11-4 用于测量光束偏转并用作液位计的 LVDT 。

电涡流接近传感器

变压器原理可用于构建接近传感器，其中次级线圈的输出电压成为其附近是否存在导电物体的敏感指示器（图TF11-5）。

►当一个物体放置在次级线圈前面时，线圈的磁场会在物体中感应出涡流（环形）电流，该涡流产生自己的磁场，其方向与次级线圈的磁场相反。◄

磁通量的减少会导致输出电压下降，变化的幅度取决于物体的导电特性及其与传感器的距离。

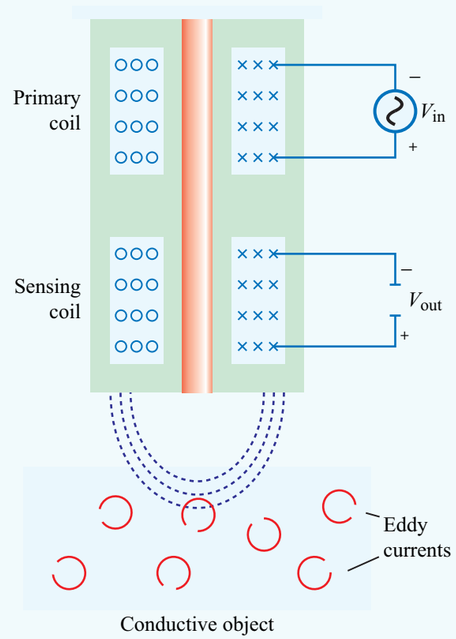


图 TF11-5 电涡流接近传感器。

Chapter 5 Summary

Concepts

- The magnetic force acting on a charged particle q moving with a velocity $\mathbf{u}$ in a region containing a magnetic flux density $\mathbf{B}$ is $\mathbf{F}_m = q\mathbf{u}\times\mathbf{B}$.
- The total electromagnetic force, known as the Lorentz force, acting on a moving charge in the presence of both electric and magnetic fields is $\mathbf{F} = q(\mathbf{E}+\mathbf{u}\times\mathbf{B})$.
- Magnetic forces acting on current loops can generate magnetic torques.
- The magnetic field intensity induced by a current element is defined by the Biot–Savart law.
- Gauss's law for magnetism states that the net magnetic flux flowing out of any closed surface is zero.
- Ampère's law states that the line integral of $\mathbf{H}$ over a closed contour is equal to the net current crossing the surface bounded by the contour.
- The vector magnetic potential $\mathbf{A}$ is related to $\mathbf{B}$ by $\mathbf{B} = \nabla\times\mathbf{A}$.
- Materials are classified as diamagnetic, paramagnetic, or ferromagnetic, depending on their crystalline structure and the behavior under the influence of an external magnetic field.
- Diamagnetic and paramagnetic materials exhibit a linear behavior between $\mathbf{B}$ and $\mathbf{H}$ with $\mu \approx \mu_0$ for both.
- Ferromagnetic materials exhibit a nonlinear hysteretic behavior between $\mathbf{B}$ and $\mathbf{H}$, and for some, μ may be as large as $10^5\mu_0$.
- At the boundary between two different media, the normal component of $\mathbf{B}$ is continuous, and the tangential components of $\mathbf{H}$ are related by $H_{2t} - H_{1t} = J_s$, where J_s is the surface current density flowing in a direction orthogonal to H_{1t} and H_{2t}.
- The inductance of a circuit is defined as the ratio of magnetic flux linking the circuit to the current flowing through it.
- Magnetic energy density is given by $w_m = \frac{1}{2}\mu H^2$.

Mathematical and Physical Models

Maxwell's Magnetostatics Equations

Gauss's Law for Magnetism

$$\nabla\cdot\mathbf{B} = 0 \quad\leftrightarrow\quad \oint_S \mathbf{B}\cdot d\mathbf{s} = 0$$

Ampère's Law

$$\nabla\times\mathbf{H} = \mathbf{J} \quad\leftrightarrow\quad \oint_C \mathbf{H}\cdot d\boldsymbol{\ell} = I$$

Lorentz Force on Charge q

$$\mathbf{F} = q(\mathbf{E}+\mathbf{u}\times\mathbf{B})$$

Magnetic Force on Wire

$$\mathbf{F}_m = I\oint_C d\mathbf{l}\times\mathbf{B} \qquad (\text{N})$$

Magnetic Torque on Loop

$$\mathbf{T} = \mathbf{m}\times\mathbf{B} \qquad (\text{N}\cdot\text{m})$$

$$\mathbf{m} = \hat{\mathbf{n}}NIA \qquad (\text{A}\cdot\text{m}^2)$$

Biot–Savart Law

$$\mathbf{H} = \frac{I}{4\pi}\int_l \frac{d\mathbf{l}\times\hat{\mathbf{R}}}{R^2} \qquad (\text{A/m})$$

Magnetic Field

Infinitely Long Wire $\quad \mathbf{B} = \hat{\boldsymbol{\phi}}\,\dfrac{\mu_0 I}{2\pi r} \qquad (\text{Wb/m}^2)$

Circular Loop $\quad \mathbf{H} = \hat{\mathbf{z}}\,\dfrac{Ia^2}{2(a^2+z^2)^{3/2}} \qquad (\text{A/m})$

Solenoid $\quad \mathbf{B} \approx \hat{\mathbf{z}}\,\mu nI = \dfrac{\hat{\mathbf{z}}\,\mu NI}{l} \qquad (\text{Wb/m}^2)$

Vector Magnetic Potential

$$\mathbf{B} = \nabla\times\mathbf{A} \qquad (\text{Wb/m}^2)$$

Vector Poisson's Equation

$$\nabla^2\mathbf{A} = -\mu\mathbf{J}$$

Inductance

$$L = \frac{\Lambda}{I} = \frac{\Phi}{I} = \frac{1}{I}\int_S \mathbf{B}\cdot d\mathbf{s} \qquad (\text{H})$$

Magnetic Energy Density

$$w_m = \frac{1}{2}\mu H^2 \qquad (\text{J/m}^3)$$

第五章总结

概念

- 在磁通密度为 B 的区域中，以速度 u 运动的带电粒子 q 受到的磁力为 $F_m = qu\times B$。
- 在存在电场和磁场的情况下作用于移动电荷的总电磁力（称为洛伦兹力）为 $F = q(E + u\times B)$。
- 作用在电流环上的磁力可以产生磁扭矩。
- 电流元件感应的磁场强度由毕奥-萨伐尔定律定义。
- 高斯磁定律指出，从任何闭合表面流出的净磁通量为零。
- 安培环路定律指出，H 在闭合轮廓上的线积分等于穿过由轮廓界定的表面的净电流。
- 矢量磁势 A 与 B 的关系为 $B = \nabla\times A$。
- 材料可分为抗磁性、顺磁性、或铁磁性，取决于它们的晶体结构和在外部磁场影响下的行为。
- 抗磁性和顺磁性材料在 B 和 H 之间表现出线性行为，两者均具有 $\mu \approx \mu_0$。
- 铁磁材料在 B 和 H 之间表现出非线性磁滞行为，对于某些材料，μ 可能大至 $10^5\mu_0$。
- B 的法向分量连续；切向磁场满足 $n\times(H_2 - H_1) = J_s$，其中 n 是从介质1指向介质2的单位法向量。
- 电感 L 为总磁链 Λ 与电流 I 之比：L=Λ/I；总磁链须计入全部匝数。
- 磁能密度由 $w_m = \frac{1}{2}\mu H^2$ 给出。

数学与物理模型

静磁场的麦克斯韦方程组

磁场的高斯定律

$$\nabla\cdot\mathbf{B} = 0 \quad\leftrightarrow\quad \oint_S \mathbf{B}\cdot d\mathbf{s} = 0$$

安培环路定律

$$\nabla\times\mathbf{H} = \mathbf{J} \quad\leftrightarrow\quad \oint_C \mathbf{H}\cdot d\boldsymbol{\ell} = I$$

电荷 q 所受洛伦兹力

$$\mathbf{F} = q(\mathbf{E}+\mathbf{u}\times\mathbf{B})$$

载流导线所受磁力

$$\mathbf{F}_m = I\oint_C d\mathbf{l}\times\mathbf{B} \qquad (\text{N})$$

电流回路所受磁力矩

$$\mathbf{T} = \mathbf{m}\times\mathbf{B} \qquad (\text{N}\cdot\text{m})$$

$$\mathbf{m} = \hat{\mathbf{n}}NIA \qquad (\text{A}\cdot\text{m}^2)$$

毕奥-萨伐尔定律

$$\mathbf{H} = \frac{I}{4\pi}\int_l \frac{d\mathbf{l}\times\hat{\mathbf{R}}}{R^2} \qquad (\text{A/m})$$

磁场

无限长直导线 $\quad \mathbf{B} = \hat{\boldsymbol{\phi}}\,\dfrac{\mu_0 I}{2\pi r} \qquad (\text{Wb/m}^2)$

圆形电流回路 $\quad \mathbf{H} = \hat{\mathbf{z}}\,\dfrac{Ia^2}{2(a^2+z^2)^{3/2}} \qquad (\text{A/m})$

螺线管 $\quad \mathbf{B} \approx \hat{\mathbf{z}}\,\mu nI = \dfrac{\hat{\mathbf{z}}\,\mu NI}{l} \qquad (\text{Wb/m}^2)$

磁矢势

$$\mathbf{B} = \nabla\times\mathbf{A} \qquad (\text{Wb/m}^2)$$

矢量泊松方程

$$\nabla^2\mathbf{A} = -\mu\mathbf{J}$$

单匝回路电感

$$L = \frac{\Lambda}{I} = \frac{\Phi}{I} = \frac{1}{I}\int_S \mathbf{B}\cdot d\mathbf{s} \qquad (\text{H})$$

磁能密度

$$w_m = \frac{1}{2}\mu H^2 \qquad (\text{J/m}^3)$$

使用条件：圆形回路的磁场式用于其轴线上；螺线管式用于足够长的螺线管内部。电感一般定义为 L=Λ/I，其中 Λ 为总磁链；只有单匝时 Λ=Φ，多匝且每匝磁通相同才有 Λ=NΦ。

Important Terms Provide definitions or explain the meaning of the following terms:

Ampère's law
Ampèrian contour
Biot–Savart law
current density (volume) **J**
diamagnetic
ferromagnetic
Gauss's law for magnetism
hard and soft ferromagnetic materials
inductance (self- and mutual)
Lorentz force **F**
magnetic dipole
magnetic energy W_m
magnetic energy density w_m
magnetic flux Φ
magnetic flux density **B**
magnetic flux linkage Λ
magnetic force $\mathbf{F}_m$
magnetic hysteresis
magnetic moment **m**
magnetic potential **A**
magnetic susceptibility χ_m
magnetization curve
magnetization vector **M**
magnetized domains
moment arm **d**
orbital and spin magnetic moments
paramagnetic
solenoid
surface current density $\mathbf{J}_s$
toroid
toroidal coil
torque **T**
vector Poisson's equation

PROBLEMS

Section 5-1: Magnetic Forces and Torques

*5.1 An electron with a speed of 8×10^6 m/s is projected along the positive x direction into a medium containing a uniform magnetic flux density $\mathbf{B} = (\hat{\mathbf{x}}4 - \hat{\mathbf{z}}3)$ T. Given that $e = 1.6 \times 10^{-19}$ C and the mass of an electron is $m_e = 9.1 \times 10^{-31}$ kg, determine the initial acceleration vector of the electron (at the moment it is projected into the medium).

5.2 When a particle with charge q and mass m is introduced into a medium with a uniform field **B** such that the initial velocity of the particle **u** is perpendicular to **B** (Fig. P5.2), the magnetic force exerted on the particle causes it to move in a circle of radius a. By equating $\mathbf{F}_m$ to the centripetal force on the particle, determine a in terms of q, m, u, and **B**.

5.3 The circuit shown in Fig. P5.3 uses two identical springs to support a 10 cm long horizontal wire with a mass of 20 g. In the absence of a magnetic field, the weight of the wire causes the springs to stretch a distance of 0.2 cm each. When a uniform magnetic field is turned on in the region containing the horizontal wire, the springs are observed to stretch an additional 0.5 cm each. What is the intensity of the magnetic flux density **B**? The force equation for a spring is $F = kd$, where k is the spring constant and d is the distance it has been stretched.

*Answer(s) available in Appendix E.

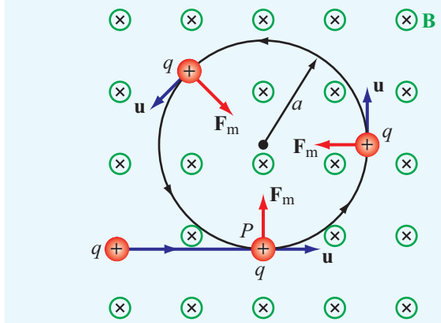


Figure P5.2 Particle of charge q projected with velocity **u** into a medium with a uniform field **B** perpendicular to **u** (Problem 5.2).

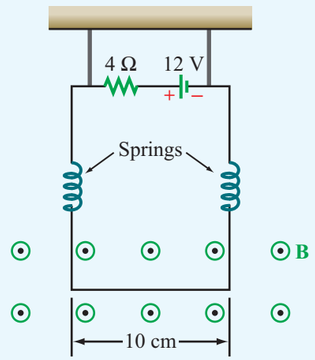


Figure P5.3 Configuration of Problem 5.3.

重要术语 提供以下术语的定义或解释其含义：

安培环路定律 安培环路 毕奥-萨伐尔定律 电流密度（体积） J 抗磁铁磁 高斯磁性定律 硬磁和软磁材料 电感（自电感和互电感）
洛伦兹力 F 磁偶极子 磁能 W_m 磁能密度 w_m 磁通 Φ 磁通密度 B 磁通链 Λ 磁力 F_m 磁滞
磁矩 m 磁势 A 磁化率 χ_m 磁化曲线 磁化矢量 M 磁畴 力矩臂 d 轨道和自旋磁矩 顺磁
螺线管表面电流密度 J_s 环形线圈 环形线圈 扭矩 T 矢量 泊松方程

问题

第 5-1 节：磁力和扭矩

*5.1 电子速度为 8×10^6 m/s 沿正 x 方向投射到含有均匀磁通密度 $B = (\hat{x}4 - \hat{z}3)$ T 的介质中。假设 $e = 1.6 \times 10^{-19}$ C 且电子质量为 $m_e = 9.1 \times 10^{-31}$ kg，确定电子的初始加速度矢量（在其投射到介质中的那一刻）。

5.2 当带电荷 q、质量 m 的粒子被引入均匀场 B 的介质中，使得粒子的初速度 u 垂直于 B 时（图 P5.2），施加在粒子上的磁力使其沿半径为 a 的圆运动。通过将 F_m 等同于粒子上的向心力，根据 q、m、u 和 B 确定 a。

5.3 图P5.3所示电路采用两个相同的弹簧支撑一根10厘米长、质量为20克的水平导线。在没有磁场的情况下，金属丝的重量会导致每个弹簧拉伸 0.2 厘米的距离。当在包含水平线的区域中打开均匀磁场时，可以观察到每个弹簧额外拉伸 0.5 厘米。磁通密度 B 的强度是多少？弹簧的力方程为 $F = kd$，其中 k 是弹簧常数，d 是弹簧拉伸的距离。

*Answer(s) available in Appendix E.

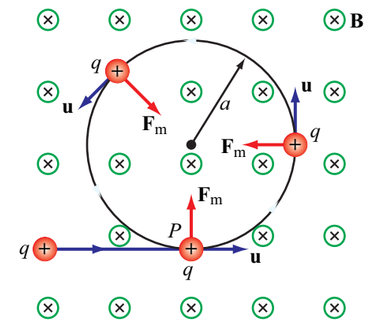


图 P5.2 电荷 q 的粒子以速度 u 投射到具有垂直于 u 的均匀场 B 的介质中（问题 5.2）。

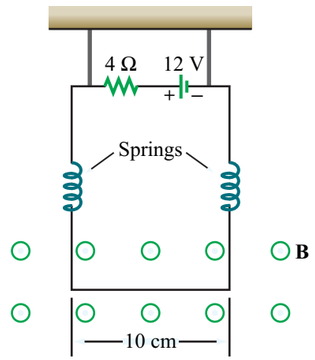


图 P5.3 问题 5.3 的配置。